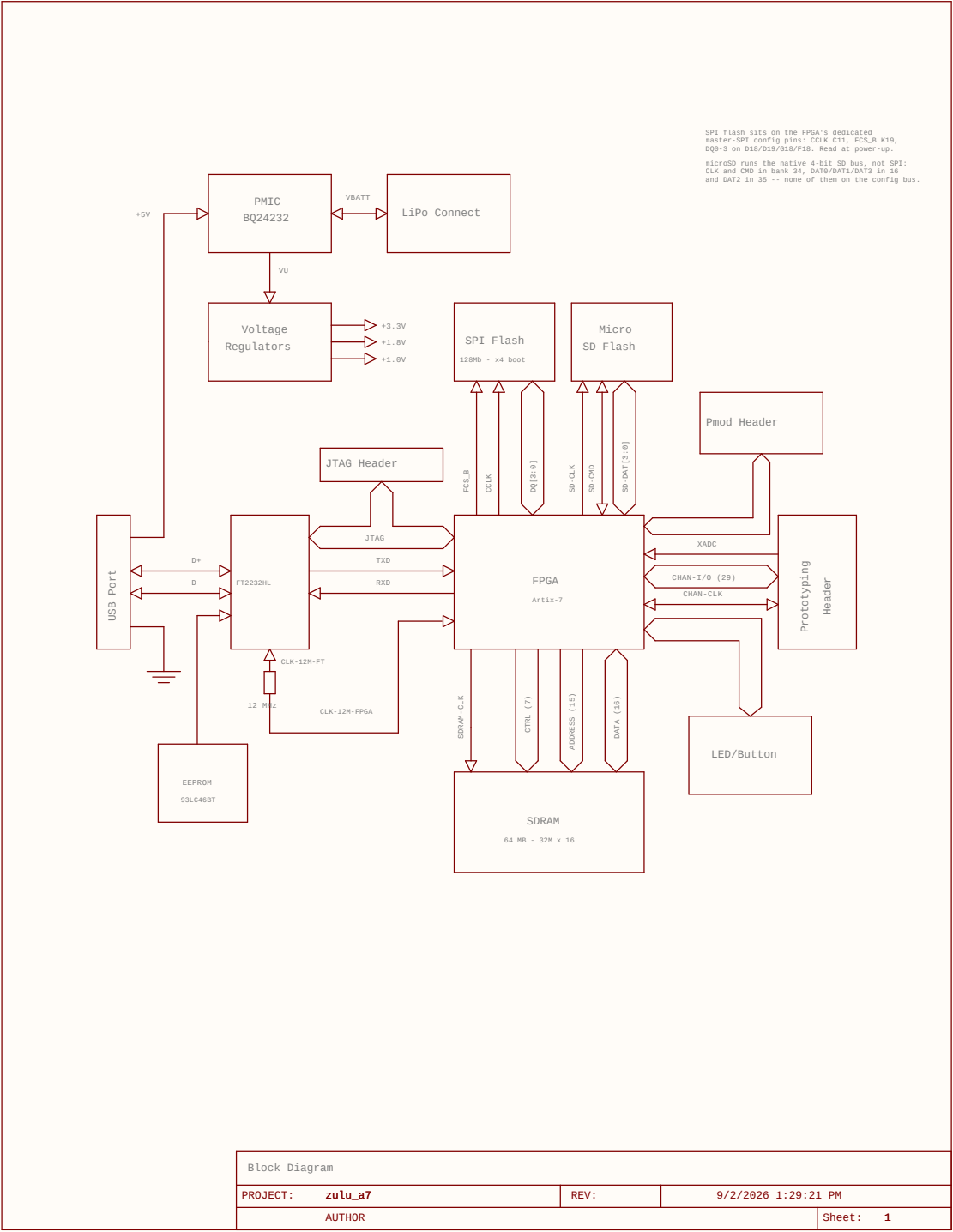
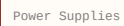
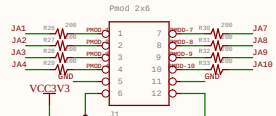
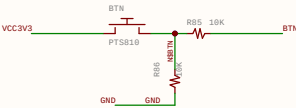
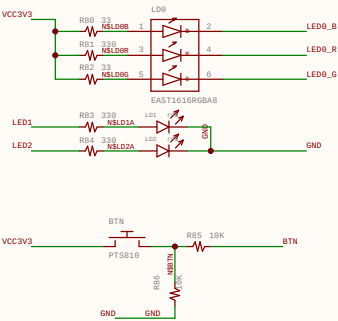
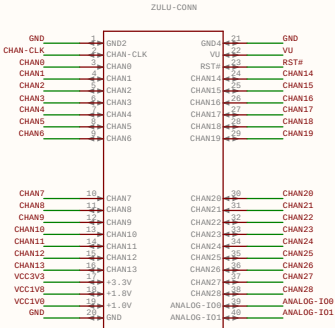


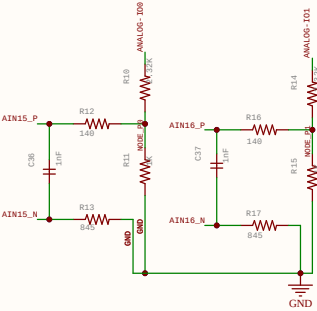




X2: 40 pins on a 2.54 mm grid, rows 22.86 mm (0.900 in) apart, numbered 1-40 straight through from the right and skipping both landings. The two rows are mirror images. TOP ROW, x 50.69 down to 1.27: GND (pin 1), CHAN-CLK (2), CHAN0-CHAN6 (3-9), then CHAN7-CHAN13 (10-16), +3.3V, +1.8V, +1.0V, GND (17-20), then CHAN14-CHAN16 (21-23), CHAN17-CHAN19 (24-26), CHAN20-CHAN28 (29-36), ANALOG-IO0, ANALOG-IO1 (37, 40). BOTTOM ROW, x 50.69 down to 1.27: GND, VU, RST* (pins 21-23), CHAN14, CHAN16 (24-26), CHAN17, CHAN19 (27-29), then CHAN20-CHAN28 (30-38), ANALOG-IO0, ANALOG-IO1 (39, 40). FOUR positions in each row -- x 36.83, 34.29, 31.75, 29.21, between pins 9 and 10 above and between pins 29 and 30 below -- are RESERVED, not spare: the edge-mounted micro-USB X1 lands on the top row (7.88 mm of copper), the top-side LiPo header X4 (JST B2B-PH-SMA-1B, 7.95 mm body) lands on the bottom row directly opposite it.



Digilent 12-pin Pmod (JA). Pins 1-4 and 7-10 are signals through 200R series resistors; pins 5/11 are GND and 6/12 are VCC3V3 -- so a 2x6 header carries EIGHT signals, not twelve, which is why the names skip JA5/JA6. No decoupling here: the Pmod spec puts bypassing on the module, and VCC3V3 already carries ~170uF. The Cmd A7 has none at its Pmod either. JA5-JA6 (the top row, the half a Pmod uses for SPI) are all in bank 16: A14 A15 B15 A16, A17 is C15, also bank 16. JA8/JA9/JA10 are W16/W2/W7 -- banks 14/35/34 -- because nothing else was free. All five banks are VCC0=3.3V, so this costs timing coherence, not levels.

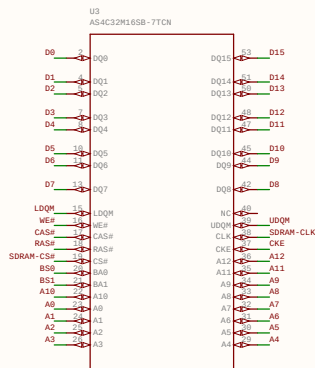


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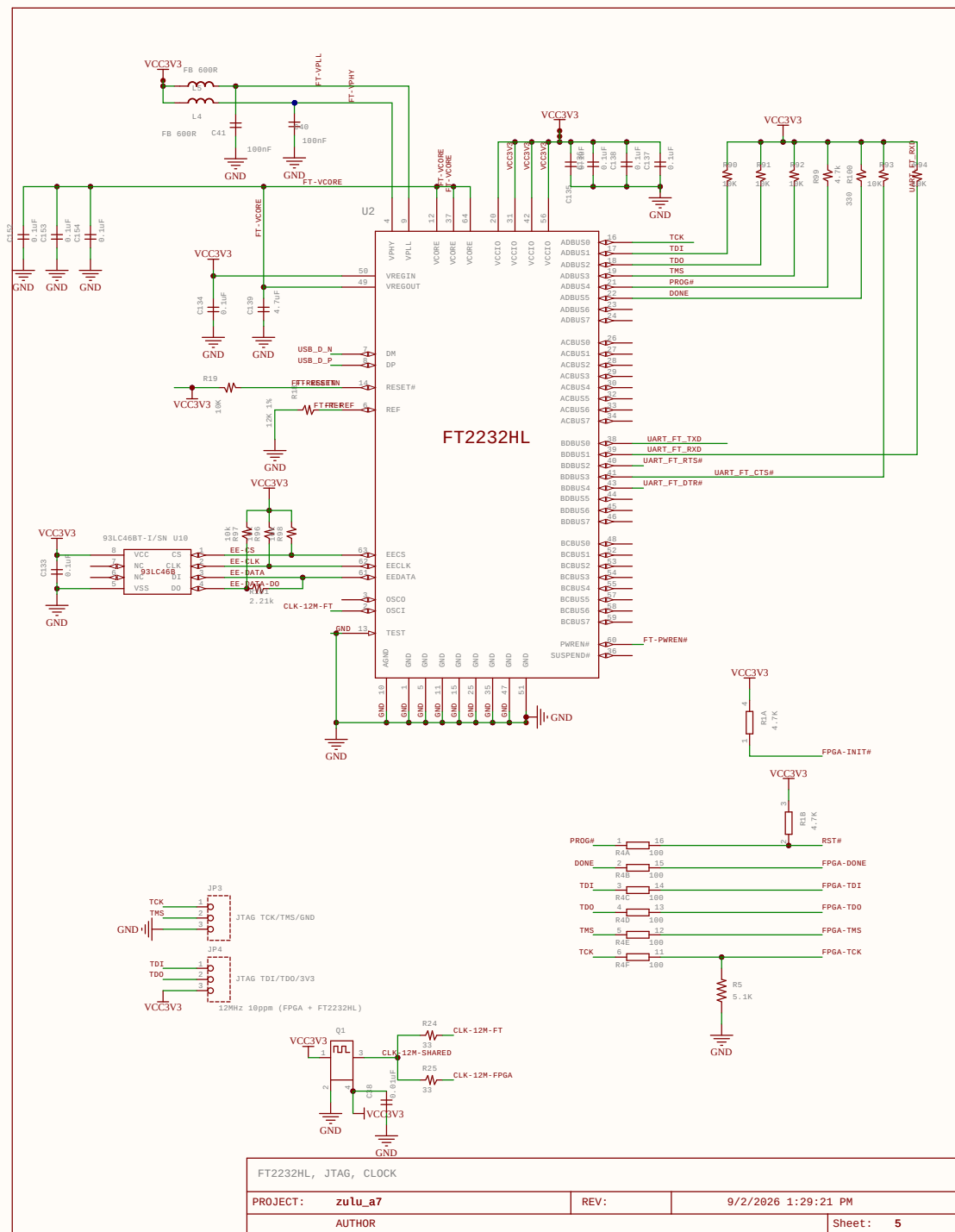
General IO			
PROJECT:	zulu_a7	REV:	9/2/2026 1:29:21 PM
AUTHOR			Sheet: 3

US = A54C32M16S8-7TCN: 512Mbit (32M x 16) = 64MB.
Pin-for-pin and dimensionally identical to the 256MB
A54C16M16SA-6TIN it replaces -- all 54 pins and every
TSDP: It's dimension checked against both datasheets, and
it is IC on the board. It is not a 16Mbit device, it's
ONE DIFFERENCE, AND IT IS NOT A PIN: the column address
is A0-A9 (1024 columns), to A0-A8 (512). The memory
controller must be set to 10 column bits, or it reaches
only half the array. Row A0-A12, banks BA0/BA1 and
refresh (8192 cycles / 64ms) are unchanged.
-7TCN since 2626-09-08 (-6TIN dry): 143 MHz max at CL3,
tRCD/tRP 21, tRC 63 ns, TA 0-70 C, tD01 max 110 ns.

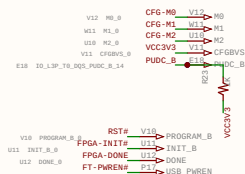


US DECOUPLING -- each cap bridges one supply pin to the ground pin OPPOSITE it.
In the 54-pin TSOP-II, pin n faces pin 55-n, and every supply pin faces a ground of its own domain: VDD 1/14/27 to VSS 54/41/28, VDDQ 3/9/43/49 to VSSQ 52/46/12/6. Alliance isolates VDDQ from VDD for DQ noise immunity -- LAYOUT MUST KEEP THE PAIRS: each cap on the underside spanning its two pins, own vias, VDDQ returning to VSSQ and VDD to VSS. C10 is bulk and pairs with nothing. Do NOT split the ground plane.

Memory			
PROJECT:	zulu_a7	REV:	9/2/2026 1:29:21 PM
AUTHOR			Sheet: 4



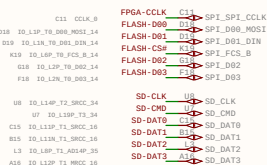
CONFIG



CLK



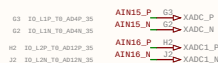
FLASH



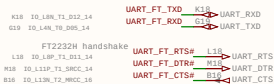
JTAG



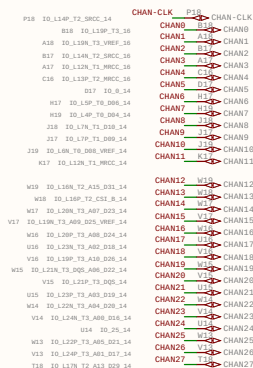
ANALOG



FT2232H



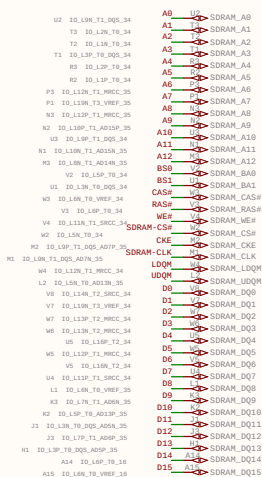
PROTOTYPING HEADER



The six pins above CCLK..D03 are the ONLY balls master-SPI configuration can use (UG470). They are ordinary user I/O once DONE goes high. SDRAM A4 moved K19 -> D17 to free FCS_B; the ESP reset moved D17 -> B15. microSD is on its own CLK/CMD/DAT0 so a card cannot fight the configuration bus.

The six microSD signals below CCLK/flash are the native 4-bit SD bus: CLK CMD DAT0-DAT3. All sit in bank 34 byte group T0 (R2 T1 T2 U1 V2 W2). DAT1/DAT2 took CHAN1/CHAN0 off the prototyping header; DAT3 is the old SPI chip-select renamed, so only two pins were spent

SDRAM



UNCONNECTED BALLS
XADC analog (unused)

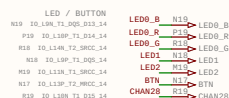
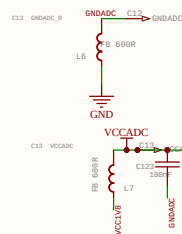


Thermal diode (unused)



Battery backup (unused)

GTP - float (UG482 T5-4)



FPGA Connections			
PROJECT:	zu1u_a7	REV:	9/2/2026 1:29:21 PM
AUTHOR			Sheet: 6

